Introduction to the Course

Lecture 00

Computer Networks and Distributed Programming

The course at a glance

University	Università degli Studî di Napoli Federico II
Degree	Bachelor's degree in Computer Science (Informatica)
Academic year	2026/27
Credits	9
Semester	I
Time of lectures	Monday 11:00am, Tuesday 2:00pm, Thursday 11:00am
Lecture rooms	A8 (Monday), A7 (Tuesday & Thursday)
Language	lectures in Italian, slides in English
Lecturer	Andrea Calí

The slides are in English because the field is: terminology, manuals, tools, and both textbooks.

What this course is about

- How the Internet works, from the applications you use every day down to the physical medium
  - all this while studying principles of networking.
- Top-down: first the applications and their protocols, then the layers underneath, one at a time.
- Along the way you write (small) network programs yourself, in C, on real sockets.
- You will look at real traffic with a packet analyser.
- We close the course with security and privacy.

A project that runs through the course

- During the course we shall analyse and build a working chat, client and server, in C.
- The project will grow through several versions, incorporating new features.

Books and material

- J. F. Kurose, K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- A. S. Tanenbaum et al., *Computer Networks*, Global Edition, Pearson, 2021.
- Adopted text: Kurose & Ross, but some topics will be covered by the other one.
- Slides and material are published on Teams and Github.
- Announcements and questions on Teams—team code `8bt07b0`.
- The slides support the lectures—do not use them as main study material.
- A board will be used during the lectures.
  - The content will be available in PDF.

Exam and office hours

Exam

Format: Written (2 h) and oral

Sessions: As per schedule

Office hours

Day: Tuesday

Time: 4:00pm–6:00pm (always by appointment, also at a different time/day)

Where: Office, via Claudio, Bldg 1

Email: andrea.cali@unina.it

The lab environment

- A ready-made Linux environment in a Docker container: compiler, network tools, packet capture, an editor.
- The same for everybody, on Linux, macOS and Windows, and it installs nothing on your machine.
- Your code stays in a folder on your own computer: the container can be thrown away and restarted clean.